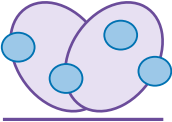
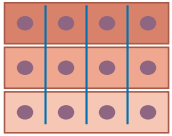
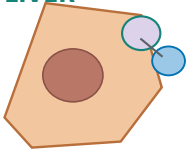
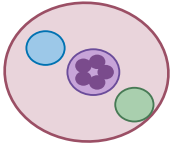


Flight-associated pathway-score patterns and tissue-state hypotheses

Tissue context	Observed lower expiMap program scores in flight	Interpretive link (not causal order)	Tissue-state hypothesis
A THYMUS 	↓ DNA repair ↓ RHOA cytoskeletal cycle ↓ Lymphoid-stromal interactions	----->	Consistent with reduced repair, cell motility, and niche coordination
B SKIN 	↓ Chromatin-modifying enzymes ↓ DNA repair ↓ Hedgehog signaling ↓ Sphingolipid metabolism ↓ Cell-cell junction organization	----->	Consistent with reduced tissue maintenance, barrier support, and cell coordination
C LIVER 	↓ MHC class II antigen presentation ↓ T-cell receptor signaling	----->	Consistent with lower adaptive immune communication
D SPLEEN 	↓ T-cell receptor signaling ↓ Neutrophil degranulation ↓ C-type lectin receptor signaling	----->	Consistent with lower adaptive activation, innate sensing, and effector-related transcription

Conceptual synthesis of bulk-tissue pathway scores. Lower scores do not prove pathway inhibition; cell-composition shifts may contribute.